

Wai Shing Tang

wstangpp@gmail.com | wtang8.github.io

Computational Scientist | Generative Models for Biological Data | Bayesian Inference | Machine Learning

RESEARCH EXPERIENCE

Microsoft Research

May 2025 –
Oct 2025

Senior Researcher, AI4Science

- Worked in an interdisciplinary AI-for-science team applying generative models to biomolecular structure prediction and design
- Integrated cryo-EM imaging data into diffusion-based generative models for protein structure
- Developed Bayesian likelihood frameworks connecting experimental structural data with generative structural models
- Gained hands-on experience with modern structural AI systems including Boltz-2 and BioEmu

Flatiron Institute

June 2022 –
April 2025

Flatiron Research Fellow, Center of Computational Mathematics (CCM) and Structural Molecular Biophysics group (SMBp)

- Led development of Bayesian inference methods for extracting structural information from extremely noisy cryo-EM imaging data
- Developed high-throughput statistical image analysis software for large-scale cryo-EM datasets
- Accelerated likelihood computations by more than 20x compared to previous implementations
- Designed HPC workflows processing terabyte-scale biological imaging datasets
- Mentored research applying Bayesian inference methods to experimental chromatin datasets to infer ensemble structural distributions

Brown University

Sept 2017 –
May 2022

Ph.D. research

Advisor: Lorin Crawford

- Developed statistical inference frameworks for detecting meaningful signals in structural and genomic-scale biological datasets
- Applied Gaussian-process feature selection models originally developed for genomics to protein structural data
- Identified topological features distinguishing protein mutants and predicted mutational effects from limited simulation datasets
- Performed large-scale molecular simulations and data analysis on HPC systems
- Developed statistical approaches to distinguish inter-class biological signal from intra-class structural noise

EDUCATION

Brown University

Sept 2017 –
May 2022

Ph.D. in Physics. Advisor: Lorin Crawford

Thesis: Topological Data Analytic framework for biological structures and Short-term evolving molecular models

The Chinese University of Hong Kong

Sept 2013 –
June 2017

B.Sc. in Physics (Enrichment Stream in Physics)

SOFTWARE

- [1] SINATRA Pro: Protein Structure and Conformation Analysis using Topological Summary Statistics (in Python and C/C++). <https://github.com/lcrawlab/SINATRA-Pro>
- [2] Ensemble reweighting using Cryo-EM particles. <https://github.com/wstangfi/Ensemble-reweighting-using-Cryo-EM-particles>
- [3] CryoLike: A Python package for Cryo-EM image-to-structure likelihood calculations <https://github.com/flatironinstitute/CryoLike>

SELECTED PUBLICATIONS

- [1] **W. Tang**, J. Soules, A. Rangan, P. Cossio (2025). CryoLike: A python package for cryo-electron microscopy image-to-structure likelihood calculations. *Acta Cryst. D*.
- [2] **W. Tang**, D.S. Sánchez, J.G. Barreto, B. Carpenter, S.M. Hanson, A.H. Barnett, E.H. Thiede, P. Cossio (2023). Ensemble reweighting using Cryo-EM particles. *Journal of Physical Chemistry B*. 127(24), 5410–5421.
- [3] **W. Tang**, E.D. Zhong, S.M. Hanson, E.H. Thiede, P. Cossio (2023). Conformational heterogeneity and probability distributions from single-particle cryo-electron microscopy. *Current Opinion in Structural Biology*. 81, 102626.
- [4] **W. Tang**, G.M. da Silva, H. Kirveslahti, E. Skeens, B. Feng, T. Sudijono, K.K. Yang, S. Mukherjee, B. Rubenstein, and L. Crawford (2022). A topological data analytic approach for discovering biophysical signatures in protein dynamics. *PLOS Computational Biology*. 18(5): e1010045.
- [5] A.C. Murthy, **W. Tang**, N. Jovic, A. Janke, D.H. Seo, T. Perdikari, J. Mittal, and N.L. Fawzi (2021). Molecular interactions contributing to FUS SYGQ LC/RGG phase separation and co-partitioning with RNA polymerase II heptads. *Nature Structural & Molecular Biology*. 28, 923–935.
- [6] **W. Tang**, N.L. Fawzi, and J. Mittal (2020). Refining All-Atom Protein Force Fields for Polar-Rich, Prion-like, Low-Complexity Intrinsically Disordered Proteins. *Journal of Physical Chemistry B*. 124(43): 9505-9512.
- [7] B.S. Schuster, G.L. Dignon, **W. Tang**, F.M. Kelley, A.K. Ranganath, C.N. Jahnke, A.G. Simpkins, R.M. Regy, D.A. Hammer, M.C. Good, and J. Mittal (2020). Identifying sequence perturbations to an intrinsically disordered protein that determine its phase-separation behavior. *PNAS*. 117(21): 11421-11431.